# Phishing Email Analyzer



## Acknowledgement


I would like to express my sincere gratitude to my teacher and for providing me with the opportunity to complete this project titled "Phishing Email Analyzer." Their valuable guidance, encouragement, and continuous support helped me understand the concepts of cybersecurity and phishing detection.

I am thankful to my institution for providing the necessary learning resources and a supportive environment that made the successful completion of this project possible

Finally, I express my gratitude to everyone who directly or indirectly contributed to the successful completion of this project. This project has enhanced my knowledge of Python programming, phishing detection techniques, and cybersecurity awareness, and has motivated me to explore more advanced cybersecurity concepts in the future.

**Project Name**
Email Phishing Detection and Analysis System

## ➢ Objective

To identify phishing emails using keyword and URL analysis.
To increase awareness about email-based cyberattacks
To help users recognize suspicious emails before opening malicious links.
To demonstrate a simple phishing detection system using Python

## ➢ Problem Statement

Phishing emails are one of the most common cyber threats. Attackers impersonate trusted organizations to steal passwords, banking information, and personal data. Many users cannot distinguish between legitimate and fake emails. This project aims to build a simple tool that analyzes email content and warns users if the email appears suspicious.

## ➢ Research Findings

Most phishing emails create urgency.
They often contain fake login links.
They use keywords like:
Verify Account

Update Password
Urgent
Click Here
Login Now
Limited Time
Many phishing emails use shortened or suspicious URLs

## ➢ Methodology

User enters email content.
The analyzer scans the email.
It searches for suspicious keywords.
It detects URLs.
It calculates a phishing score.
Displays Safe, Suspicious, or Highly Suspicious

## ➢ Tools Used

Python 3
VS Code / Pydroid 3
Regular Expressions (Regex)

## ➢ Implementation

**Features**

Detect phishing keywords
Detect URLs
Calculate risk score
Display warning message

## ➢ Source code Python

```
import re
phishing_keywords = [
    "urgent",
    "verify",
    "password",
    "bank",
```

```
    "login",
    "click here",
    "limited time",
    "confirm",
    "security alert",
    "account suspended",
    "update account"
]
def analyze_email(email):
    score = 0
    found_keywords = []
    email_lower = email.lower()
    for word in phishing_keywords:
        if word in email_lower:
            score += 1
            found_keywords.append(word)
    urls = re.findall(r'https?://\S+|www\.\S+', email)
    score += len(urls)
    print("\n------ Analysis Result ------")
    if found_keywords:
        print("Suspicious Keywords:")
        for k in found_keywords:
            print("-", k)
    if urls:
        print("\nURLs Found:")
        for u in urls:
            print("-", u)
    print("\nRisk Score:", score)
    if score >= 5:
        print("HIGHLY SUSPICIOUS EMAIL")
    elif score >= 2:
        print("SUSPICIOUS EMAIL")
    else:
        print("SAFE EMAIL")
```

```
email = input("Paste Email Content:\n\n")
analyze_email(email)
```

## ➢ Sample input

Dear User,
Your account has been suspended.
Click here:
http://fake-bank-login.com
Verify your password immediately to avoid account closure

## ➢ Sample output

Suspicious Keywords:
- verify
- password
- account suspended
- click here

**URLs Found:**
http://fake-bank-login.com
Risk Score: 5
HIGHLY SUSPICIOUS EMAIL

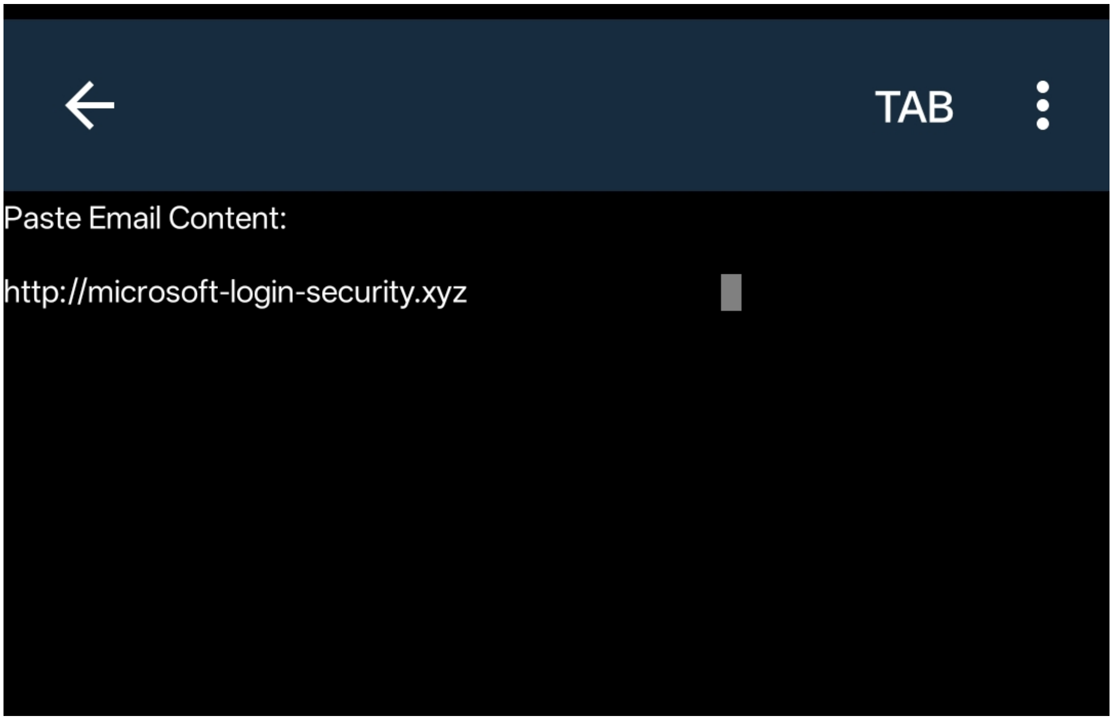
# Screenshot

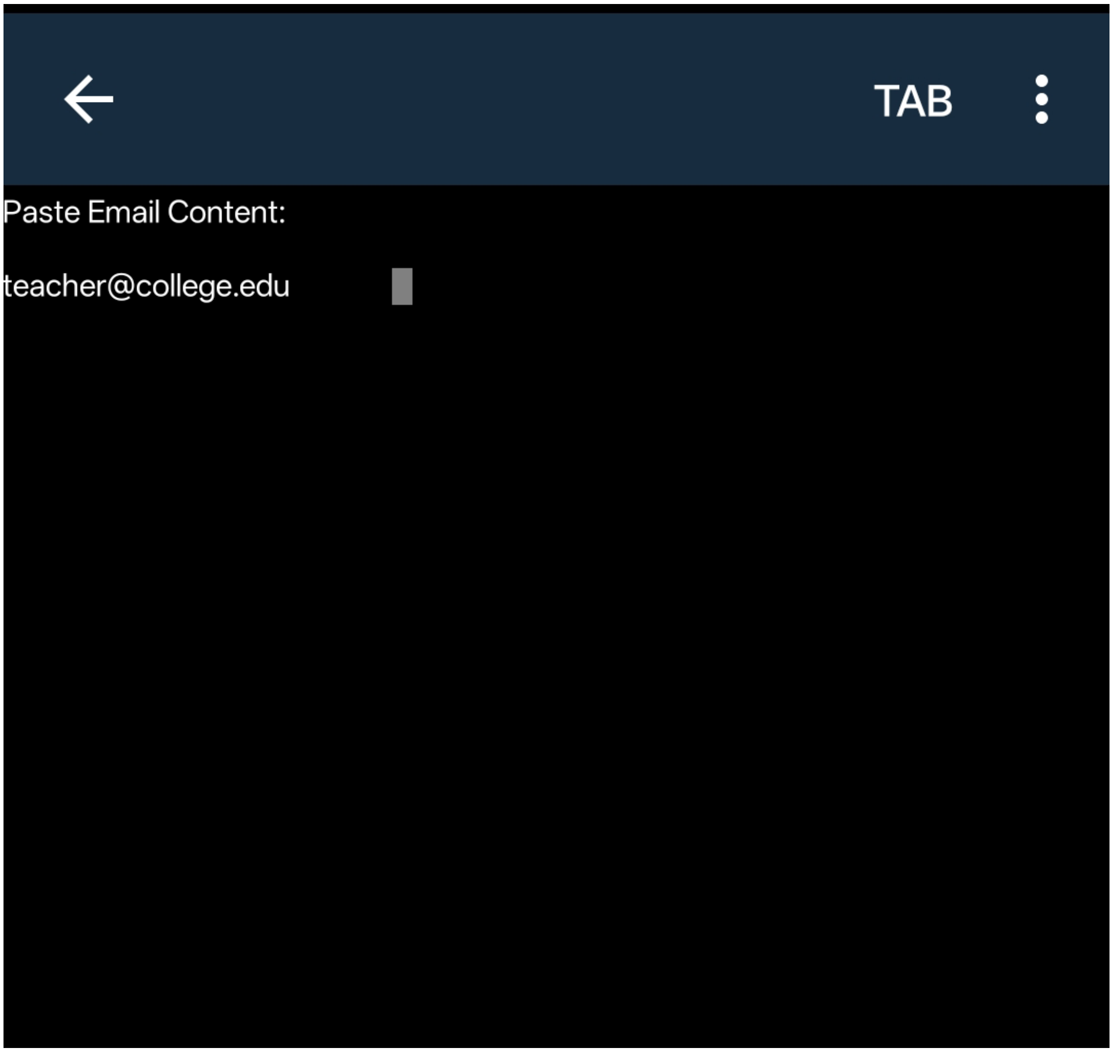
### Project Code and running the program

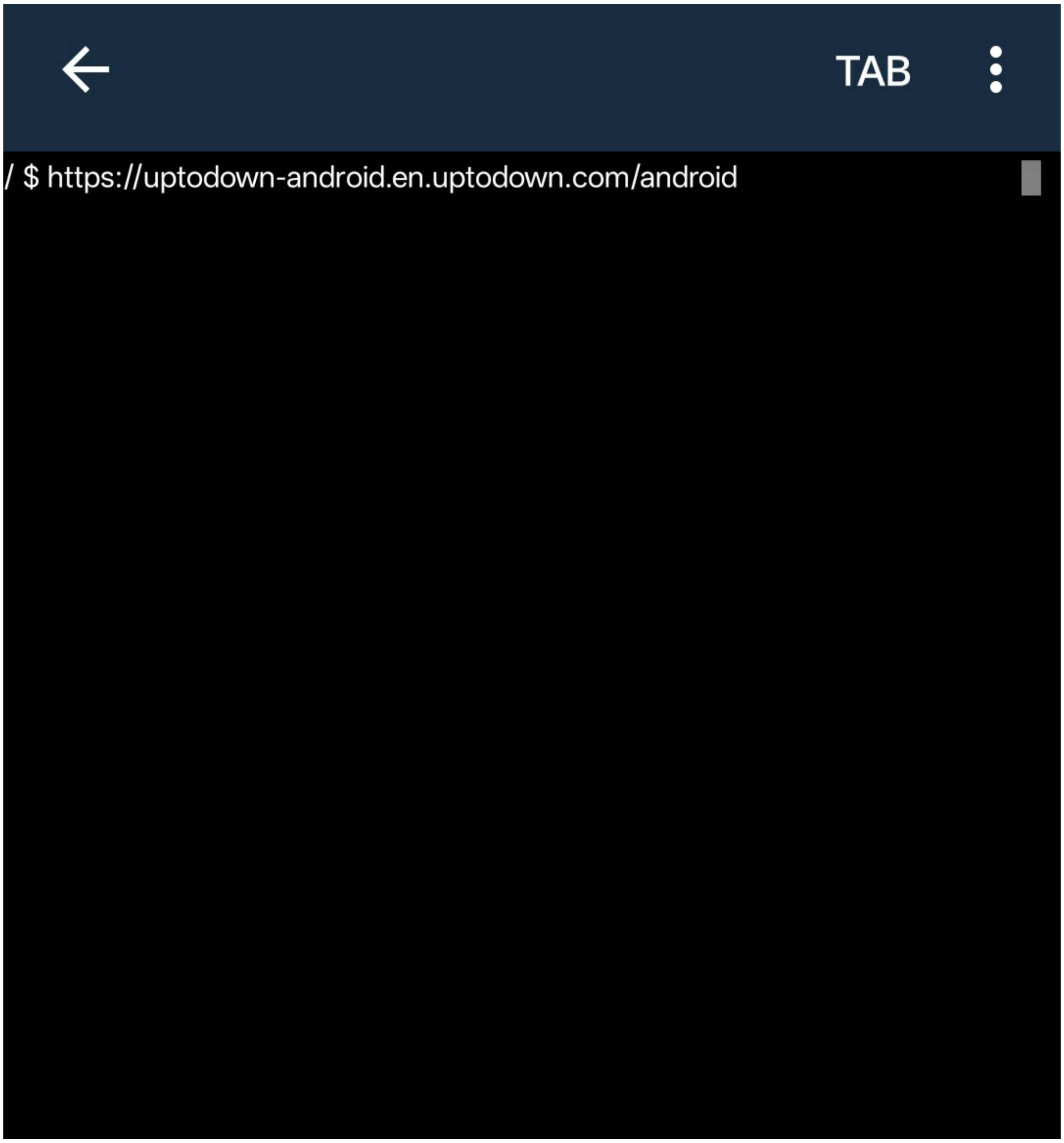
```
import re

phishing_keywords = [
    "urgent",
    "verify",
    "password",
    "bank",
    "login",
    "click here",
    "limited time",
    "confirm",
    "security alert",
    "account suspended",
    "update account"
]

def analyze_email(email):

    score = 0
    found_keywords = []

    email_lower = email.lower()

    for word in phishing_keywords:
        if word in email_lower:
            score += 1
            found_keywords.append(word)

    urls = re.findall(r'https?://\S+|www\.\S+', email)

    score += len(urls)

    print("\n------ Analysis Result
```

```python

    score += len(urls)

    print("\n------ Analysis Result ------")

    if found_keywords:
        print("Suspicious Keywords:")
        for k in found_keywords:
            print("-", k)

    if urls:
        print("\nURLs Found:")
        for u in urls:
            print("-", u)

    print("\nRisk Score:", score)

    if score >= 5:
        print("HIGHLY SUSPICIOUS EMAIL")
    elif score >= 2:
        print("SUSPICIOUS EMAIL")
    else:
        print("SAFE EMAIL")


email = input("Paste Email Content:\n\n")

analyze_email(email)
```

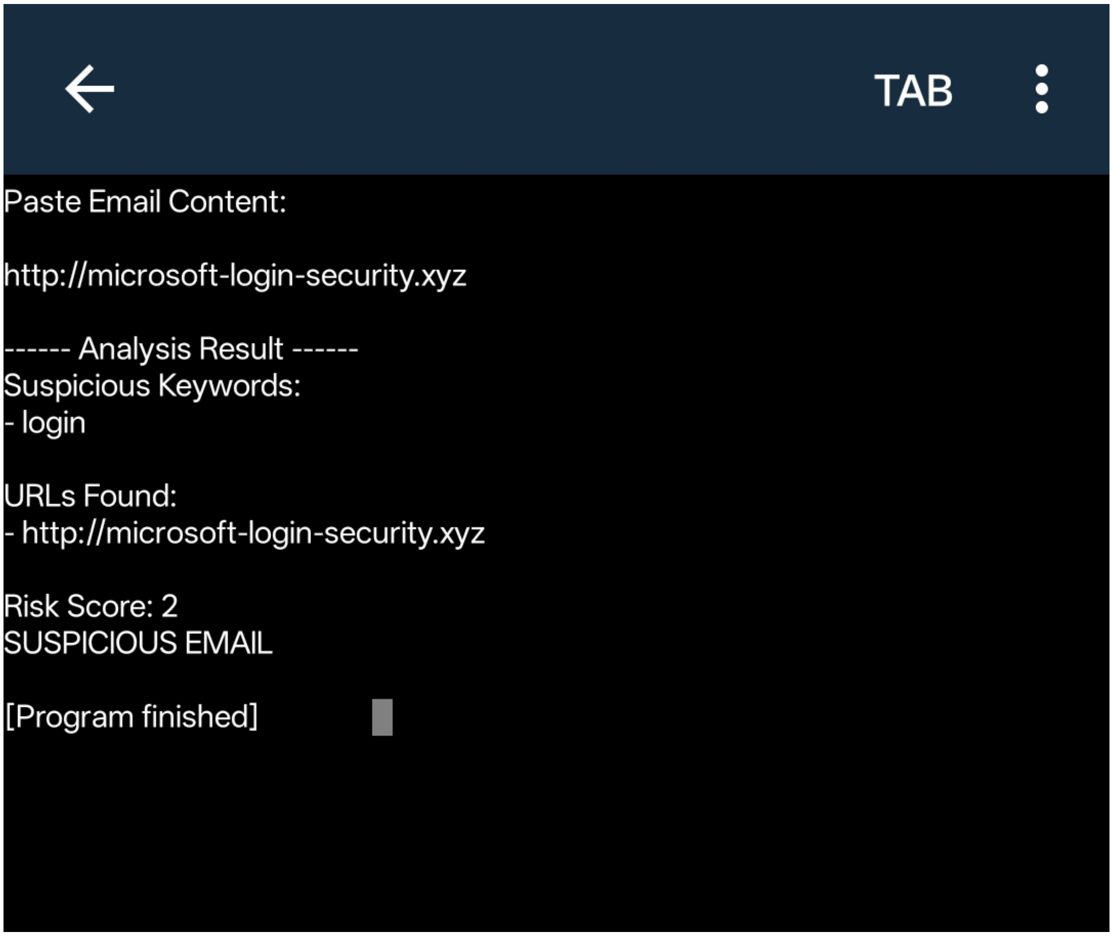
## Sample phishing email input

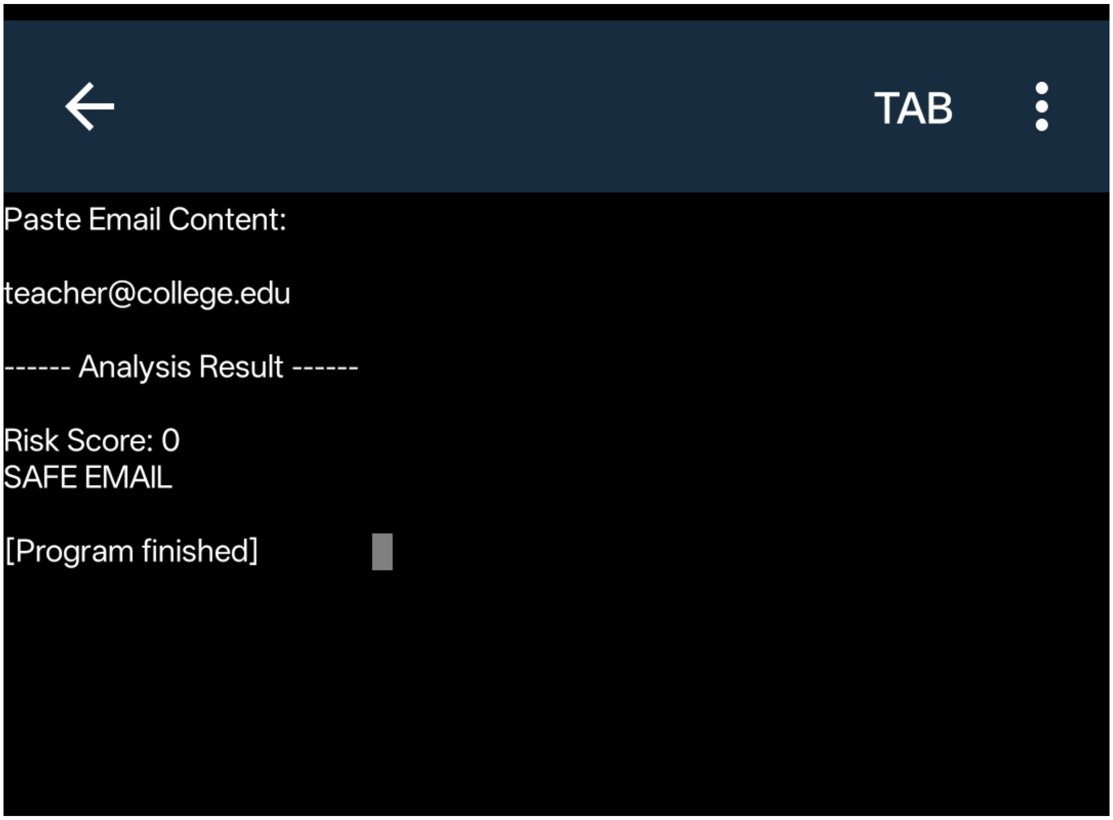


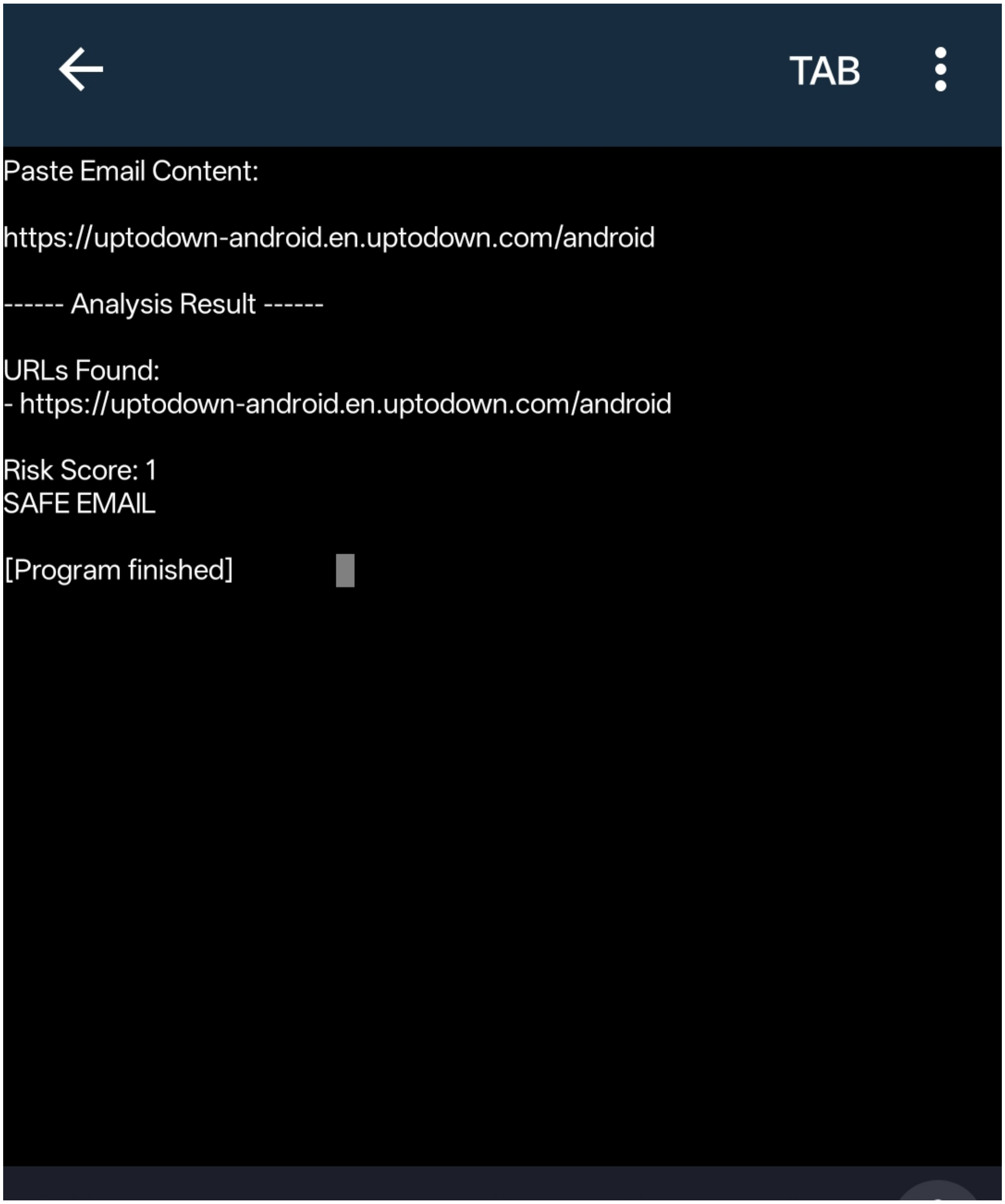

TAB
Paste Email Content:
teacher@college.edu



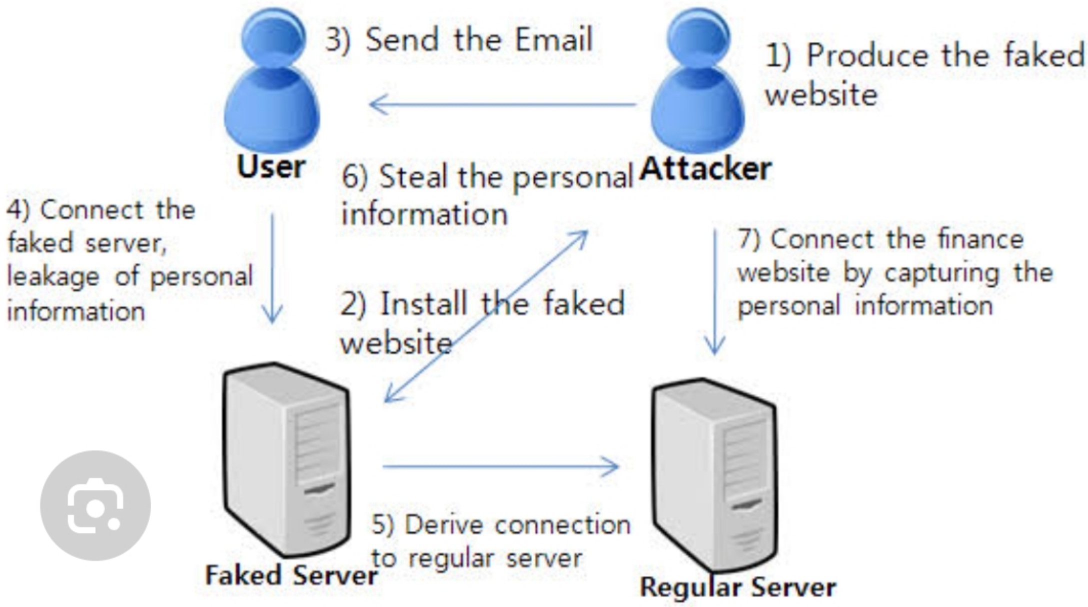

## Detection result output


TAB
Paste Email Content:
http://microsoft-login-security.xyz
------ Analysis Result ------
Suspicious Keywords:
- login
URLs Found:
- http://microsoft-login-security.xyz
Risk Score: 2
SUSPICIOUS EMAIL
[Program finished]


TAB
Paste Email Content:
teacher@college.edu
------ Analysis Result ------
Risk Score: 0
SAFE EMAIL
[Program finished]


TAB
Paste Email Content:
https://uptodown-android.en.uptodown.com/android
------ Analysis Result ------
URLs Found:
- https://uptodown-android.en.uptodown.com/android
Risk Score: 1
SAFE EMAIL
[Program finished]

# Flowchart




## Show Risk Level

Replace your existing result section with:

```
if score >= 5:
   result_box.insert(tk.END, "   HIGHLY SUSPICIOUS EMAIL")
elif score >= 2:
   result_box.insert(tk.END, "   SUSPICIOUS EMAIL")
else:
   result_box.insert(tk.END, "   SAFE EMAIL")
```

## ⮟ Results

Successfully detects suspicious keywords.
Identifies URLs in email content.
Calculates phishing risk.
Displays warning messages based on risk level.

## ⮟ Challenges

Cannot detect advanced phishing techniques.
Uses keyword matching instead of machine learning.
May generate false positives.

## ➢ Future Scope

Machine Learning–based detection.
Integration with email clients.
Real-time email scanning.
QR code phishing detection.
AI-powered phishing analysis.
URL reputation checking.
Attachment scanning.

## Conclusion

The Phishing Email Analyzer is a simple cybersecurity project that demonstrates how suspicious email content can be analyzed using Python. Although it uses basic keyword and URL detection, it helps users understand common phishing indicators and promotes cybersecurity awareness. With future enhancements such as machine learning and real-time scanning, it can become a more effective phishing detection tool.